

FILE : CH11EG1

```
t=()          # it is an empty tuple
print(t)
t=(3)         # it is an integer not a tuple
print(t)
t=3,          # it is a tuple
print(t)
t=(3,)        # it is also a tuple
print(t)

#2
print()
print("#2")
t1=tuple("hello") # it is a tuple
print(t1)

#3
print()
print("#3")
tuple=eval(input("enter a tuple to be added:"))
print("Tuple you entered: ",tuple)
```

FILE : CH11EG2

```
t1=eval(input("Enter a input for tuple 1 : "))
t2=eval(input("Enter a input for tuple 2 : "))
t3=eval(input("Enter a input for tuple 3 : "))
print("Type of t1 : ",type(t1))
print("Type of t2 :",type(t2))
print("Type of t3 :",type(t3))
t1=tuple(t1)
t2=tuple(t2)
t3=tuple(t3)
print("Tuple1 ",t1)
print("Tuple2 ",t2)
print("Tuple3 ",t3)
print("Type of t1 : ",type(t1))
print("Type of t2 : ",type(t2))
print("Type of t3 : ",type(t3))
```

FILE : CH11EG3

```
t1=tuple(eval(input("Enter a input: ")))  
print("create tuple1 is:",t1)
```

FILE : CH11EG4

```
t1=eval(input("Enter a input: "))  
print("type 1 : ",type(t1))
```

FILE : CH11EG5

```
t1=eval(input("Enter a input: "))  
print(t1[0],t1[-1])  
print(t1[1],t1[-2])  
print(t1[2],t1[-3])  
print(t1[2-3]," = ",t1[-1])
```

```
numbers=([1,2,3],[4,5,6])  
#numbers[1]=[1,2,0]  
numbers[0][2]=0  
print(numbers[0][2])  
print(numbers)
```

FILE : CH11EG6

```
T=('p','y','t','h','o','n')  
print(type(T))  
for a in T :  
    print(T[a])
```

FILE : CH11EG7

```
T=('Hello','Isnt','python','fun','?')  
length=len(T)  
for a in range(length):  
    print("at index ",a,"and",(a-length),"element:",T[a])
```

FILE : CH11EG8

```
tp11=(1,3,5)
tp12=(6,7,8)
print(tp11+tp12)
print(tp11+(3,))
print(tp11*3)
```

FILE : CH11EG9

```
tup=eval(input("enter input for tuple: "))
t=tup[3:-3]
t0=tup[3:tup[5]]
t1=tup[::-3]
t2=tup[2:8:2]
t3=tup[3:30]
print("two created tuple are:")
print("Tuple:",t)
print("Tuple0:",t0)
print("Tuple1:",t1)
print("Tuple2:",t2)
print("Tuple3:",t3)
print(tup[2:5]*3)
print(tup[2:5]+(3,4))
```

FILE : CH11EG10

```
tup=eval(input("enter input for tuple: "))
t1=tup[2:8:2]
t2=tup[-3:-9:-2]
print("two created tuple are:")
print("Tuple1:",t1)
print("Tuple2:",t2)
```

FILE : CH11EG11

```
tup=eval(input("enter input for tuple: "))
t1=tup[2:8:2]
t2=tup[-3:-9:-2]
t3=t2[::-1]
if t3==t1:
    print("the two tuple contain the same element in reverse
order.")
else:
    print("the two tuple contain different element.")
```

FILE : CH11EG12

```
tup=eval(input("enter a 4 element tuple:"))
a,b,c,d=tup
print("tuple unpacked in ",a,b,c,d)
tup=c,d,a,b
print("Tuple after swapping the element:",tup)
```

FILE : CH11EG13

```
tup=eval(input("enter a tuple: "))
mn=min(tup)
ma=max(tup)
print("minimum element",mn,"is at index ",tup.index(mn))
print("minimum element",ma,"is at index ",tup.index(ma))
print("sum of elements of tuple(tup)",sum(tup))
```

FILE : CH11EG14

```
tup=eval(input("Enter a tuple: "))
ln=len(tup)
num=tup.count(tup[0])    # counts the no. of times a particular
element
print(num)               # is present in the tuple 'tup'
if num==ln:
    print("Tuple contains all the same element ")
else:
    print("Tuple contains different element")
```

FILE : CH11EG15

```
print("To print an empty tuple" ,tuple())
print("To print a tuple" ,tuple("abc"))
print("To print a tuple from a list: " ,tuple([1,2,3]))
print("To print a tuple from a dictionary: "
,tuple({1:"1",2:"2",3:"3"}))
print("To print a list from a tuple of a list : "
,list[tuple([1,2,3])]) # list converted to tuple using tuple
object
```

and back again to list using list object

```
t1=eval(input("Enter a tuple: "))
lst=sorted(t1)
t1=tuple(lst)
print("Tuple after sorting:",t1)
```

FILE : CH11EG16

```
t1=(11,21,31,42,51)
t11=list(t1)
t11[-2]=41
t1=tuple(t11)
print("modified tuple: ",t1)
```

FILE : CH11EG17

```
student=(11,'Rai',(67,77,78,82,80))
print("minimum marks obtained:",min(student[2]))
print("maximum marks obtained:",max(student[2]))
print("total marks obtained:",sum(student[2]))
```

FILE : CH11EG18

```
student=(11,'Ria',(66,77,78,82,80))
print("minimum marks obtains",min(student[2]))
print("maximum marks obtains",max(student[2]))
print("total marks obtains",sum(student[2]))
```